



उत्पाद मानुयल  
पैकेजबंद प्राकृतिक जल और पैकेजबंद पेय जल के लिए प्लास्टिक की बोतलें/ कंटेनर - विशिष्टि  
IS 15410:2003 के अनुसार

**PRODUCT MANUAL FOR  
PLASTICS BOTTLES /CONTAINERS FOR PACKAGED NATURAL MINERAL WATER AND  
PACKAGED DRINKING WATER ACCORDING TO IS 15410:2003**

भारतीय मानक ब्यूरो (अनुरूपता मूल्यांकन) विनियम 2018 की स्कीम-I के तहत यह उत्पाद मानुयल प्रमाणीकरणके प्रचलन में रीति और पारदर्शिता के सुसंगत सुनिश्चित करने के लिए सभी क्षेत्रीय/शाखा कार्यालयों एवं लाइसेन्स धारियों द्वारा संदर्भ सामग्री के रूप में उपयोग किया जाएगा। बीआईएस लाइसेन्स/प्रमाण पत्र प्राप्त करने के इच्छुक भावी आवेदकों द्वारा भी इस दस्तावेज़ का उपयोग किया जा सकता है।

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure coherence of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification licence/certificate.

|    |                                                      |   |                                                                                                                                                                                                                                               |
|----|------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | उत्पाद<br><b>Product</b>                             | : | IS 15410:2003                                                                                                                                                                                                                                 |
|    | शीर्षक<br><b>Title</b>                               | : | पैकेजबंद खनिज जल और पैकेजबंद पेय जल की पैकेजबंदी के लिए धारक- विशिष्टि<br>Plastics Bottles /Containers for Packaged Natural Mineral Water and Packaged Drinking Water - Specification                                                         |
|    | संशोधन संख्या<br><b>No. of Amendments</b>            | : | 07                                                                                                                                                                                                                                            |
| 2. | नमूनाकरण दिशा निर्देश<br><b>Sampling Guidelines:</b> |   |                                                                                                                                                                                                                                               |
| a) | कच्चा माल<br><b>Raw material</b>                     | : | a) Manufacturer to submit test report/ test certificate in compliance to Clause 4.1 of IS 15410:2003.<br>b) Manufacturer to submit declaration/ undertaking in compliance to Clause 4.2.1 of IS 15410:2003 for the material used for top lids |
| b) | समूहिकरण दिशा निर्देश<br><b>Grouping guidelines</b>  | : | कृपया Annex-ए देखें<br>Please see Annex – A                                                                                                                                                                                                   |
| c) | नमूने का परिमाण                                      | : | 10 Filled + 6 Empty + 30 Caps for each type (Bottle, Jar,                                                                                                                                                                                     |

|    |                                                                                                                                                                                                                                                        |   |                                                                                                                                                                                       |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Sample Size</b>                                                                                                                                                                                                                                     |   | glasses, cups) of containers<br>(Declaration of wall thickness shall be obtained during the drawl of sample)                                                                          |
| 3. | परीक्षण उपकरणों की सूची<br><b>List of Test Equipment</b>                                                                                                                                                                                               | : | कृपया Annex-B देखें<br>Please see Annex – B                                                                                                                                           |
| 4. | निरीक्षण व परीक्षण स्कीम<br><b>Scheme of Inspection and Testing</b>                                                                                                                                                                                    | : | कृपया Annex-C देखें<br>Please see Annex – C                                                                                                                                           |
| 5. | <b>एक दिन में संभावित परीक्षण</b><br><b>Possible tests in a day:</b>                                                                                                                                                                                   |   |                                                                                                                                                                                       |
|    | i.Design, Shape and Dimensions<br>ii.Manufacture, Workmanship, Finish and Appearance<br>iii.Capacity(maybe done for all nominal capacities)<br>iv.Wall thickness<br>v.Transparency<br>vi.Closure Leakage<br>vii.Air pressure leakage<br>viii.Drop test |   |                                                                                                                                                                                       |
| 6. | <b>लाइसेंस का कार्यक्षेत्र / Scope of the Licence:</b>                                                                                                                                                                                                 |   |                                                                                                                                                                                       |
|    | <b>IS 15410:2003</b> के अनुसार मानक मुहर का उपयोग करने के लिए लाइसेंस निम्नलिखित कार्यक्षेत्रके लिए प्रदान किया जाता है<br>“Licence is granted to use Standard Mark as per 15410:2003 with the following scope:                                        |   |                                                                                                                                                                                       |
|    | उत्पाद का नाम<br>Name of the product                                                                                                                                                                                                                   |   | पैकेजबंद खनिज जल और पैकेजबंद पेय जल की पैकेजबंदी के लिए धारक- विशिष्टि<br>Plastics Bottles /Containers for Packaged Natural Mineral Water and Packaged Drinking Water - Specification |
|    | <b>Material</b>                                                                                                                                                                                                                                        |   | PE/PVC/PET/PBT/PP/Polycarbonate/Polystyrene                                                                                                                                           |
|    | <b>Type of container</b>                                                                                                                                                                                                                               |   | Bottle, Jar, glasses, cups etc                                                                                                                                                        |
|    | <b>Nominal Capacity</b>                                                                                                                                                                                                                                |   | In ml or litres (pack sizes)                                                                                                                                                          |

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**ANNEX – A**  
**TO THE PRODUCT MANUAL FOR PLASTICS BOTTLES / CONTAINERS FOR PACKAGED**  
**NATURAL MINERAL WATER AND PACKAGED DRINKING WATER**  
**ACCORDING TO IS 15410:2003**  
**GROUPING GUIDELINES**

1. The manufacturer shall declare the material, type of container, nominal capacity of container and wall thickness(Top, Middle and Bottom) of container which they intend to cover in their scope of licence.
2. The sample of each type of material (*PE/PVC/PET/PBT/PP/Polycarbonate/Polystyrene*) and each type of container (*Bottle, Jar, glasses, cups etc*) shall be subjected to testing for considering grant of licence/inclusion of that type. However, sample of the highest nominal capacity of each type of material and each type of container may be subjected to testing for considering grant of licence/inclusion of that material and type of container, for all pack sizes.  
  
Further, test requirement for capacity( as per Clause 4.4 of IS 15410:2003) may be ensured through in-house test report for all pack sizes, to be considered under the scope of licence.
3. It shall be ensured that the firm is having all the necessary manufacturing and testing facilities for the types to be included in the licence.
4. During the operation of licence, BO shall ensure that all the nominal capacities of Plastics Bottles / Containers for Packaged Natural Mineral Water and Packaged Drinking Water covered in the license are drawn for independent testing on rotation over a period of time.

**ANNEX – B**  
**TO THE PRODUCT MANUAL FOR PLASTICS BOTTLES / CONTAINERS FOR PACKAGED  
NATURAL MINERAL WATER AND PACKAGED DRINKING WATER  
ACCORDING TO IS 15410:2003**

**List of test equipment**

**Major test equipment required to test as per the Indian Standard**

| Sr. No. | Tests used in with Clause Reference  | Test equipment                                                                                                                                                                                                                                                                                                                                    |
|---------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.      | Capacity<br>Cl. 4.4                  | i) Suitable plastic disc to cover the neck face of the container<br>ii) Suitable weighing balance to an accuracy of 0.1g<br>iii) Equipments to maintain temperature $27 \pm 2^\circ\text{C}$<br>(or)<br>Suitable measuring cylinders                                                                                                              |
| 2.      | Wall Thickness<br>Cl. 4.5            | Ball ended micrometer or dial caliper gauge fitted with spherical anvils, to an accuracy of 0.02mm                                                                                                                                                                                                                                                |
| 3.      | Transparency<br>Cl. 4.6.2            | Integration ball type light transmittance measurement device as described in Annex A of IS 15410.                                                                                                                                                                                                                                                 |
| 4.      | Leakage Test<br>Cl. 4.6.3            | (a) <u>Closure leakage test</u><br>i. Blotting paper<br>ii. Stop watch<br>(b) <u>Vibration Leakage Test</u><br>(i) Device /methods meeting conditions mentioned in Cl.6.2.1 of IS 2798<br>(c) <u>Air Pressure Leakage Test</u><br>(i) Device /methods meeting conditions mentioned in Cl.6.3.2 of IS 2798                                         |
| 5.      | Drop Test<br>Cl. 4.6.4               | (i) Equipment /methods meeting conditions mentioned in Cl.8.2 of IS 2798<br>(ii) Scale to measure a height of 0.5m/1.0m/1.2m                                                                                                                                                                                                                      |
| 6.      | Migration Test<br>Cl. 4.6.5          | i) Electric oven/Water bath equipped with temperature controller to maintain desired temperature up to $\pm 1^\circ\text{C}$ accuracy<br>ii) Electric hot plate with temperature control regulator<br>iii) Weighing balance with a sensitivity of 0.1mg<br>iv) Beaker<br>v) Evaporating Dish<br>vi) Suitable scale /caliper to measure dimensions |
| 7.      | Water Potability test<br>Cl. 4.6.6   | Equipments to maintain temperature $38 \pm 2^\circ\text{C}$                                                                                                                                                                                                                                                                                       |
| 8.      | Specific Migration test<br>Cl. 4.6.7 | Electric oven/water bath equipped with thermostat, Electric hot plate with temperature control regulator, Analytical balance, Glass beakers, Stainless steel evaporating dish, Stainless steel tongs                                                                                                                                              |

*The above list is indicative only and may not be treated as exhaustive*

**ANNEX C**  
**TO PRODUCT MANUAL FOR**  
**PLASTICS BOTTLES /CONTAINERS FOR PACKAGED NATURAL MINERAL WATER**  
**AND PACKAGED DRINKING WATER**  
**ACCORDING TO IS 15410:2003**

**Scheme of Inspection And Testing**

**1. LABORATORY** - A laboratory shall be maintained which shall be suitably equipped (as per the requirement given in column 2 of Table 1) and staffed, where different tests given in the specification shall be carried out in accordance with the methods given in the specification.

**1.1** The manufacturer shall prepare and implement a calibration plan for the test equipments.

**2. TEST RECORDS** – The manufacturer shall maintain test records for the tests carried out to establish conformity.

**3. PACKING AND MARKING-** –The Standard Mark, as given in the Schedule of the licence, shall be marked on each container provided always that the product so marked conform to requirements of the specification.

**3.1 Packing** - The material shall be packed in suitable form of packing as agreed between the purchaser and the supplier, in line with the provisions of Clause 5.1 of IS 15410:2003.

**3.2 Marking** - The containers shall bear legibly and indelibly the information as mentioned under clause 5.2 of IS 15410:2003. In addition, the following details shall be mentioned on each container:

- i) BIS Licence No. (CM/L.....)
- ii) For details of BIS certification details please visit [www.bis.gov.in](http://www.bis.gov.in).

**4. CONTROL UNIT** – For the purpose of this scheme all the containers of the same type of material, same design/shape manufactured in one shift and manufactured from same consignment of raw material shall constitute a Control Unit.

**5. LEVELS OF CONTROL** – The tests as indicated in column 1 of Table 1 and at the levels of control specified in column 3 of Table 1, shall be carried out on the whole production of the factory which is covered by this plan and appropriate records maintained in accordance with paragraph 2 above.

**5.1** All the production which conforms to the Indian Standards and covered by the licence should be marked with Standard Mark.

**6. REJECTIONS** –Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act, 2016.

**TABLE 1 (Levels of Control)**  
(Para 5 of Scheme of Inspection and Testing)

| (1)          |                                               |                            |                   | (2)                                                                                  | (3)               |                      |                                                                                                                                                                                             |
|--------------|-----------------------------------------------|----------------------------|-------------------|--------------------------------------------------------------------------------------|-------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Details |                                               |                            |                   | Test equipment<br>requirement<br>R: required (or)<br>S: Sub-contracting<br>permitted | Levels of Control |                      |                                                                                                                                                                                             |
| Cl.          | Requirement                                   | Test<br>Method<br>Cl. Ref. | Test Method<br>IS |                                                                                      | No. of<br>Sample  | Frequency            | Remarks                                                                                                                                                                                     |
| 4.1          | Material                                      | 4.1                        | IS 15410          | S                                                                                    | One               | Every<br>consignment | The sample may be got tested in an approved outside lab. In case the consignment is accompanied with a test certificate from manufacturer, no testing shall be required.                    |
| 4.2          | Design Shapes & Dimensions                    | 4.2                        | IS 15410          | R                                                                                    | One               | Every half an hour   | See Note-3 below                                                                                                                                                                            |
| 4.3.2        | Manufacture, workmanship, finish & appearance | 4.3                        | IS 15410          | R                                                                                    | ----              | Each container       | See Note-3 below                                                                                                                                                                            |
| 4.4          | Capacity                                      | 5                          | IS 2798           | R                                                                                    | One               | Every hour           | Five consecutive samples shall be tested and afterwards one sample every hour for each machine. In case of failure, all the containers from the preceding hour production shall be checked. |
| 4.5          | Wall thickness                                | 4.5                        | IS 2798           | R                                                                                    | One               | Every hour           | Five consecutive samples shall be tested and afterwards one sample every hour for each machine. In case                                                                                     |

|       |                |         |          |   |     |                |                                                                                                                                                                                                                                                                   |
|-------|----------------|---------|----------|---|-----|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                |         |          |   |     |                | of failure, all the containers from the preceding hour production shall be checked.                                                                                                                                                                               |
| 4.6.2 | Transparency   | Annex A | IS 15410 | R | One | Every hour     | Five consecutive samples shall be tested and afterwards one sample every hour for each machine. In case of failure, all the containers from the preceding hour production shall be checked.                                                                       |
| 4.6.3 | Leakage test   | 6       | IS 2798  | R | One | Every hour     | Five consecutive samples from every hour production to be tested. For this every hour production is to be kept separate. In case of failure another two samples to be tested, in case another failure is observed, the whole lot is rejected.                     |
| 4.6.4 | Drop test      | 8       | IS 2798  | R | One | Every 4 hours  | - do -                                                                                                                                                                                                                                                            |
| 4.6.5 | Migration test | 6       | IS 9845  | R | One | Every 24 hours | One sample from every four hour production to be tested. For this every 4-hour production is to be kept separate. In case of failure another two samples to be tested, in case another failure is observed, the whole lot to be rejected. Separate tests shall be |

|       |                    |        |          |   |     |             |                             |
|-------|--------------------|--------|----------|---|-----|-------------|-----------------------------|
|       |                    |        |          |   |     |             | done for different mixture. |
| 4.6.6 | Water potability   | AnnexB | IS 15410 | S | 6   | Every month | -                           |
| 4.6.7 | Specific Migration | ---    | IS 9845  | S | One | Every month |                             |

Note-1: Whether test equipment is required or sub-contracting is permitted in column 2 shall be decided by the Bureau and shall be mandatory. Sub-contracting is permitted to a laboratory recognized by the Bureau or Government laboratories empanelled by the Bureau.

Note-2: Levels of control given in column 3 are only recommendatory in nature. The manufacturer may define the control unit/batch/lot and submit his own levels of control in column 3 with proper justification for approval by BO Head.

Note -3: A sample shall be taken at the defined frequency, which shall be examined visually for design, shape and dimensions. Each container shall also be examined visually to be free from any cavities, crevices, flaws, stains. If the sample does not conform to the specification in any one or more of these requirements, the material manufactured during the prior to drawal of sample shall either be rejected or reprocessed for its conformity to these requirements of the specification.